

Human Development Index (HDI) Predictor

Scalability & Future Plan

Project: Human Development Index (HDI) Predictor

Phase: Project Demonstration

1. Current System Limitations

- Runs locally only — no public URL without additional deployment
- Trained on a 144-country generated dataset rather than the full official UNDP dataset
- Uses only 4 indicators, while the official HDI formula also includes Expected Years of Schooling
- No user authentication or prediction history storage

2. Scalability Plan

Area	Current State	Scalable Solution
Deployment	Local Flask dev server	Deploy to Render / AWS / Google Cloud for public access
Dataset	Generated 144-country dataset	Swap in the full official Kaggle/UNDP dataset
Web Server	Flask development server	Gunicorn or Waitress for production traffic
Features	4 indicators	Add Expected Years of Schooling to match the full UNDP formula

3. Future Feature Roadmap

Priority	Feature	Description
High	Real dataset integration	Replace the generated dataset with the official UNDP/Kaggle HDI data
High	Cloud deployment	Deploy to Render or AWS for a public demo URL
Medium	Batch prediction	Upload a CSV of multiple countries and download HDI predictions
Low	Historical trend view	Show a country's HDI trend over multiple years

4. Long-Term Vision

The long-term vision is to evolve this project into a production-grade development analytics tool that researchers and policymakers can use to explore real-time, indicator-driven HDI estimates and comparisons across countries.